Pokhara University
Faculty of Science and Technology

Course No.: Elective	Full marks: 100
Course title: **Generative Artificial Intelligence (3-1-2)**	Pass marks: 45
Nature of the course: Theory/Tutorial/Practical	Time per period: 1 hour
Year:	Total periods: 45
Level: Undergraduate	Program: BE Computer/IT/Software

Course Description

This course offers an in-depth introduction to Generative Artificial Intelligence (GenAI), combining foundational programming skills with cutting-edge developments in large language models and AI application frameworks. Students will begin by mastering Python and essential data libraries, then delve into the architecture of LLMs, advanced prompting techniques, and tools like LangChain for building complex generative systems. The course also covers Retrieval Augmented Generation (RAG), ethical considerations in AI, and emerging topics such as multimodal and agentic systems. By the end of the course, students will be equipped to design, build, and critically evaluate GenAI applications across a range of contexts.

General Objectives

The course is designed with the following objectives:

1. Build foundational skills in Python and essential data libraries for AI development.
2. Understand the architecture, training, and evolution of large language models in generative AI.
3. Apply prompt engineering, LangChain, and RAG techniques to build and optimize GenAI applications.
4. Analyze ethical, social, and practical implications of GenAI in real-world contexts.

Methods of Instruction

Contents in detail with specific objectives

Specific Objectives	Contents
Develop foundational Python skills and familiarize with core data libraries and tools useful for generative AI development.	**Unit 1: Introduction to Generative AI and Python Fundamentals (7 hrs)** 1.1 Evolution of AI and Emergence of Generative AI 1.2 Python Syntax and Data Types 1.3 Control Structures and Functions in Python 1.4 Introduction to NumPy for Numerical Operations 1.5 Pandas for Data Manipulation and Analysis 1.6 Data Visualization with Matplotlib 1.7 Introduction to FastAPI for API development
Familiarize with the architecture, types, and training methodologies of modern large language models.	**Unit 2: Foundations of Large Language Models (6 hrs)** 2.1 Transformer Architecture and Attention Mechanisms 2.2 Key LLM Types: GPT, BERT, LLaMA, 2.3 Training Methodologies: Pre-training and Fine-tuning
Implement the zero-shot and few shot prompts to build effective Gen AI applications.	**Unit 3: Prompt Engineering Fundamentals (8 hrs)** 3.1 Introduction to Prompt Engineering 3.2 Anatomy of Effective Prompts 3.3 Zero-shot Learning Techniques 3.4 Few-shot Learning and Examples 3.5 Chain-of-Thought Prompting

	3.6 Role Prompting and System Instructions 3.7 Prompt Templates and Frameworks 3.8 Case Study: Interactive Prompt Workshop with LLM models
Build complex LLM-based applications using the LangChain framework by leveraging chains, memory, and tool integration.	**Unit 4: LangChain Framework and Applications (8 hrs)** 4.1 LangChain Architecture and Components 4.2 Chains and Sequence Processing 4.3 Prompt Templates and LLM Integration 4.4 Memory Systems in LangChain 4.5 Agents and Tools Implementation 4.6 Document Loading and Text Splitting 4.7 Building Conversational Applications 4.8 Case Studies of LangChain Applications
Implement context-aware LLM systems using vector embeddings, similarity search, and evaluation metrics in RAG pipelines.	**Unit 5: Retrieval Augmented Generation (RAG) (8 hrs)** 5.1 Introduction to Retrieval Augmented Generation 5.2 Understanding Tokens and Token Limitations 5.3 Vector Embeddings: Principles and Generation 5.4 Vector Databases: Pinecone, Chroma, 5.5 Chunking Strategies and Metadata 5.6 Similarity Search and Retrieval Mechanisms 5.7 Hybrid Search Methods 5.8 Re-ranking Techniques 5.9 Evaluation Metrics: BLEU, Context precision, recall, faithfulness, relevance
Analyze ethical challenges in generative AI and apply responsible AI practices across the model lifecycle.	**Unit 6: Ethical and Responsible AI (4 hrs)** 6.1 Introduction to Ethics in Generative AI 6.2 Sources and Impacts of Bias in LLMs 6.3 Fairness, Representation, and Social Harm 6.4 Explainability and Transparency Challenges

	6.5 Model Misuse, Hallucinations, and Jailbreak Risks 6.6 Data Privacy, Consent, and Legal Considerations (e.g., GDPR, AI Act)
Familiarize students with emerging trends in generative AI, including multimodal systems, agentic models, and advanced fine-tuning techniques.	**Unit 7: Advanced Topics in Generative AI (4 hrs)** 7.1 Fundamentals of Multimodal AI 7.2 Agentic AI Systems 7.3 Emerging Research Directions 7.4 Self-hosting LLMs 7.5 Finetuning LLMs

List of Tutorials

The following tutorial activities of 15 hours per group of maximum 24 students should be conducted to cover all the required contents of this course.

1. Python Environment Setup & Basics: Installing Python, Jupyter, and essential libraries; basic syntax and data types practice.
2. NumPy & Pandas Fundamentals: Core operations with NumPy arrays and Pandas DataFrames for AI data preparation.
3. Data Visualization & FastAPI: Creating visualizations with Matplotlib and building simple APIs with FastAPI.
4. LLM API Integration: Making calls to popular LLM APIs and handling responses.
5. Zero-shot Prompting: Crafting and testing effective zero-shot prompts across various tasks.
6. Few-shot Prompting: Implementing few-shot learning techniques with examples.
7. Chain-of-Thought Prompting: Developing prompts for complex reasoning tasks.
8. Role Prompting & System Instructions: Creating specialized prompts for different contexts and roles.
9. Prompt Templates & Frameworks: Building reusable prompt structures and formats.
10. LangChain Basics: Setting up LangChain environment and implementing basic chains.
11. LangChain Memory Systems: Implementing different memory types for contextual conversations.
12. LangChain Agents & Tools: Building agents that can use tools to solve multi-step problems.

13. RAG Document Processing: Techniques for processing, chunking, and preparing documents.
14. Vector Embeddings & Databases: Working with embeddings and configuring vector databases.
15. RAG Pipeline Implementation: Building and testing complete RAG systems.

Practical Work

1. Python Data Science Portfolio (5 hours)
    a. Comprehensive data analysis project using NumPy, Pandas, and Matplotlib
    b. Creation of reusable data processing functions for AI applications
    c. Integration with FastAPI to create a simple data service
2. Prompt Engineering Showcase (6 hours)
    a. Develop and document a collection of optimized prompts using a single LLM
    b. Implement zero-shot, few-shot, and chain-of-thought approaches
3. Conversational AI Assistant (6 hours)
    a. Develop a command-line AI assistant using LLM APIs
    b. Implement context management and conversation history
    c. Add domain-specific capabilities through prompt engineering
4. Advanced LangChain Application (7 hours)
    a. Create a full-featured application using LangChain framework
    b. Implement chains, agents, memory systems, and tool integration
5. Comprehensive RAG System (6 hours)
    a. Develop an end-to-end RAG pipeline for domain-specific documents
    b. Implement chunking strategies and similarity search
    c. Create a simple interface for querying the knowledge base

Evaluation system and students' responsibilities

Internal Evaluation

In addition to the formal end-semester exam(s), the internal (formative) evaluation of a student may consist of quizzes, assignments, lab reports, projects, class participation and presentation etc. The tabular presentation of the internal evaluation is as follows. The components may differ according to the nature of the subjects.

Internal Evaluation	**Weight**	**Marks**	**External Evaluation**	**Marks**
Theory				
Attendance & Class Participation				
Assignments				
Presentations/Quizzes				
Internal Assessment				
Practical				
Attendance & Class Participation				
Lab Report/Project Report				
Practical Exam/Project Work				
Viva				
Total Internal				

Student requirements:

Each student must secure at least 45% marks in internal evaluation with 80% attendance in the class in order to appear in the semester-end examination. Failing to get such a score will be equated with NOT QUALIFIED (NQ) and the student will not be eligible to appear in the End- Semester examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. Failure of a student to attend a formal exam, quiz, test, etc. won't qualify him/her for re-exam. ***Students are required to complete all the requirements defined for the completion of the course***

Prescribed Books and References

1. Prompt Engineering Guide by West, H., & Bommasani, R. (2023). GitHub. https://github.com/dair-ai/Prompt-Engineering-Guide
2. Language models are few-shot learners by Brown, T. B., et al. (2020). arXiv preprint arXiv:2005.14165. https://arxiv.org/abs/2005.14165
3. Retrieval-augmented generation for knowledge-intensive NLP tasks by Lewis, P., et al. (2020). arXiv preprint arXiv:2005.11401. https://arxiv.org/abs/2005.11401
4. LLaMA: Open and efficient foundation language models by Touvron, H., et al. (2023). arXiv preprint arXiv:2302.13971. https://arxiv.org/abs/2302.13971
5. Python for Data Analysis: Data Wrangling with pandas, NumPy, and Jupyter by McKinney, W. (2022). https://wesmckinney.com/book/
6. FastAPI Cookbook: Develop high-performance APIs and web applications with Python (2023)
7. The Big Book of Generative AI by Databricks (2025).
8. Rothman, D. (2024). RAG-Driven Generative AI: Build custom retrieval augmented generation pipelines with LlamaIndex, Deep Lake, and Pinecone. Packt.